Dallas International School

Test 8

Integration

Calculator Allowed

12th IB – Math AA

For SL & HL Students

Name : . . . . . .

EXERCISE 1 Find :

1. $\int \left(e^{7-3x}\right) dx$ [2]

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2. $\int \frac{1}{2x-1}\, dx$ [2]

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EXERCISE 2 Integrate by substitution :

1. $\int 4x^3(2+x^4)^3\, dx$ [3]

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2. $\int \sqrt{\sin x}\cos x\, dx$ [3]

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EXERCISE 3 Evaluate using technology :

$\int_1^3 \ln x\, dx$ [2]

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EXERCISE 4 [5]

The derivative of a function g is given by $g'(x) = \cos x + e^{2x}$, where $x \in \mathbb{R}$.

Given that $g(0) = 7$, find $g(x)$.

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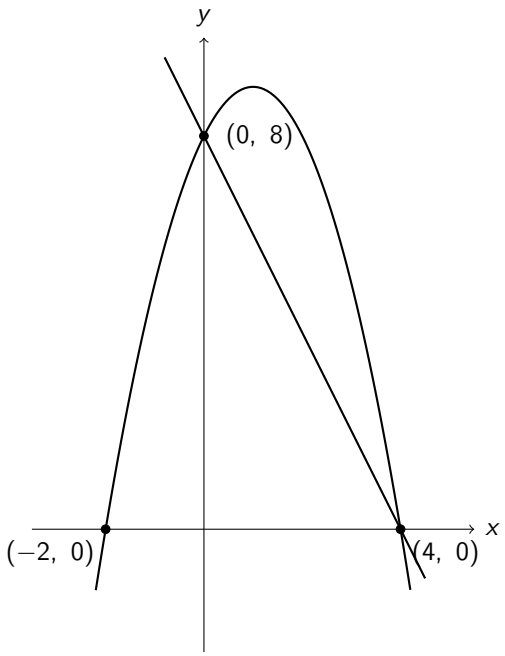
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EXERCISE 5 [16]

The following diagram shows parts of the graphs of two functions f and g.

The graph of f is linear, has an x-intercept at $(4,\ 0)$ and a y-intercept at $(0,\ 8)$.

The graph of g has x-intercepts at $(-2,\ 0)$ and $(4,\ 0)$ and a y-intercept at $(0,\ 8)$.



1. Write down the equation for f in the form $f(x) = mx + c$. [2]

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The function g is given by $g(x) = -x^2 + bx + 8$, where b is a real constant.

2. Find the value of b. [3]

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3. Show that the area of the region enclosed by the graph of f and the graph of g can be represented by the definite integral

$$\int_0^4 \left(-x^2 + 4x\right) \mathrm{d}x.$$

[2]

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4. Hence, find the area of the region enclosed by the graph of f and the graph of g. [4]

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Point P is on the graph of g. The tangent to the graph of g at P is parallel to the graph of f.

5. Find the coordinates of P. [5]

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Answer Key

Exercise 1

1. $e^x + \frac{5}{2}e^{2x} + C$
2. $\frac{3}{5}e^{5x-2} + C$
3. $-\frac{1}{3}e^{7-3x} + C$
4. $\frac{1}{2}\ln|2x-1| + C$
5. $-\frac{5}{3}\ln|1-3x| + C$
6. $-e^{-x} - 2\ln|2x+1| + C$

Exercise 2

1. $\frac{1}{8}(2+x^4)^4 + C$
2. $\frac{2}{3}(\sin x)^{3/2} + C$
3. ≈ 1.30 (by technology)

Exercise 3

$g(x) = \sin x + \frac{1}{2}e^{2x} + \frac{13}{2}$

Exercise 4

1. $f(x) = -2x + 8$
2. $b = 2$
3. (verification/show question)
4. Area $= \frac{32}{3}$ square units
5. $\mathrm{P} = (2,\ 12)$